

# **CUSTOMER CHURN ANALYSIS USING POWER BI**

## **Future Interns – Data Science & Analytics Internship**

### **Task 2: Customer Churn Analysis Dashboard**

Prepared By

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### **Introduction**

Customer retention has become one of the most important factors influencing business success in highly competitive industries. Acquiring a new customer generally costs significantly more than retaining an existing one. Therefore, understanding customer behaviour and identifying the reasons behind customer churn enables organizations to develop effective retention strategies and improve long-term profitability.

Customer churn refers to the percentage of customers who discontinue using a company's products or services during a specific period. High churn rates often indicate customer dissatisfaction, increasing competition, pricing issues, or service quality concerns. By analyzing customer data, organizations can identify common characteristics of customers who are likely to leave and take preventive actions before churn occurs.

In this project, customer information from Telco company was analyzed using Microsoft Excel and Microsoft Power BI. The dataset was cleaned, transformed, and visualized to develop an interactive dashboard capable of identifying churn patterns across different customer segments. The dashboard enables decision-makers to monitor key performance indicators and supports data-driven business decisions aimed at improving customer retention.

### **Problem Statement**

Customer churn represents a major challenge for subscription-based businesses because it directly affects revenue, profitability, and customer lifetime value. Although companies collect large volumes of customer information, deriving meaningful insights from raw data can be difficult without proper analysis.

The organization requires an interactive dashboard that can:

- Monitor customer churn.
- Identify customer groups with high churn.
- Understand the influence of contracts, payment methods, and internet services on customer retention.
- Support strategic business decisions through data visualization.

### **Objectives**

The primary objectives of this project are:

- To analyze customer churn behaviour using customer demographic and service information.

- To clean and prepare the dataset for analysis.
- To build an interactive dashboard using Microsoft Power BI.
- To calculate important business KPIs using DAX.
- To identify factors contributing to customer churn.
- To provide meaningful business insights and practical recommendations for improving customer retention.

## Dataset Description

The dataset contains customer information collected from a Telco company.

It includes both customer demographic information and subscription-related details.

### Dataset Size

- Total Records: **7,043**
- Total Columns: **21**

### Important Attributes

| Attribute       | Description                                        |
|-----------------|----------------------------------------------------|
| customerID      | Unique customer identifier                         |
| Gender          | Male or Female                                     |
| SeniorCitizen   | Indicates whether the customer is a senior citizen |
| Partner         | Whether the customer has a partner                 |
| Dependents      | Whether the customer has dependents                |
| Tenure          | Number of months with the company                  |
| PhoneService    | Phone subscription status                          |
| InternetService | Internet service type                              |
| Contract        | Contract duration                                  |
| PaymentMethod   | Customer payment method                            |
| MonthlyCharges  | Monthly bill amount                                |
| TotalCharges    | Total amount charged                               |
| Churn           | Customer retention status                          |

The target variable in this project is **Churn**, which indicates whether a customer has left the company.

## Tools and Technologies Used

The following tools were used during the project:

- **Microsoft Excel**

Used for:

- Data cleaning
- Missing value identification
- Data validation
- Preparing the dataset before visualization

- **Microsoft Power BI**

Used for:

- Data modelling
- DAX calculations
- Interactive dashboard development
- KPI creation
- Data visualization

- **Data Cleaning and Preparation**

Before beginning the analysis, the dataset was carefully examined to ensure that it was suitable for business reporting.

The following data preparation steps were performed:

- **Duplicate Record Validation**

The dataset was checked for duplicate customer records using Microsoft Excel.

**Result:**

No duplicate records were identified.

- **Missing Value Identification**

The dataset was examined for missing values.

During the analysis, **11 blank values** were identified in the **TotalCharges** column.

These records were investigated further.

It was observed that all 11 customers had a **tenure of 0 months**, indicating that they had recently joined the company and had not yet completed their first billing cycle.

Instead of deleting these records, the blank values were replaced with **0**, preserving valid customer information while maintaining data consistency.

- **Data Validation**

The following numeric fields were validated:

- Tenure
- Monthly Charges
- Total Charges

No negative values were identified.

Therefore, no additional corrections were required.

## **DAX Measures Created**

Several DAX measures were created to calculate important business KPIs.

| <b>Measure</b>          | <b>Purpose</b>                     |
|-------------------------|------------------------------------|
| Total Customers         | Counts total customers             |
| Churned Customers       | Counts customers who left          |
| Active Customers        | Calculates retained customers      |
| Churn Rate %            | Calculates churn percentage        |
| Average Monthly Charges | Calculates average monthly billing |
| Average Tenure          | Calculates average customer tenure |

These measures enable dynamic calculations that automatically respond to dashboard filters.

## **Dashboard Development**

An interactive dashboard was developed to provide a comprehensive overview of customer churn.

The dashboard contains:

### **KPI Cards**

- Total Customers
- Churn Rate
- Churned Customers
- Active Customers
- Average Monthly Charges
- Average Tenure

### **Interactive Filters**

Users can filter dashboard results by:

- Contract

- Internet Service
- Payment Method
- Gender

## Visualizations

- Churned Customers by Contract
- Churned Customers by Payment Method
- Churned Customers by Internet Service
- Customers by Gender
- Churned Customers by Senior Citizen Status
- Customer Distribution by Contract Type

These visualizations allow business users to quickly identify customer segments with higher churn rates.

## Dashboard Explanation

Each visualization was selected to answer a specific business question.

- KPI Cards provide an overview of customer retention performance.
- Contract analysis identifies which contract types experience the highest churn.
- Internet Service analysis highlights service categories with higher customer attrition.
- Payment Method analysis helps determine whether billing preferences influence churn.
- Gender distribution provides demographic information.
- Senior Citizen analysis compares churn between senior and non-senior customers.
- Contract Distribution visualization compares retained and churned customers across different contract types.

## Business Insights

- The company currently serves approximately **7,043 customers**, with **1,869 customers having churned**, resulting in an overall churn rate of **27%**.
- This indicates that nearly one out of every four customers has discontinued the service.
- Customers subscribed to **month-to-month contracts** exhibit substantially higher churn compared to customers with one-year and two-year contracts.
- Long-term contracts appear to contribute positively to customer retention.
- Customers using **Fiber Optic internet service** account for the highest number of churned customers.
- This may indicate service quality concerns, pricing issues, or unmet customer expectations within this service category.

- Customers making payments through **Electronic Check** demonstrate the highest churn among all payment methods.
- This suggests that payment experience or customer profile may influence customer retention.
- The company retains approximately **5,174 active customers**, indicating that while overall retention remains satisfactory, focused retention initiatives targeting high-risk customer groups could significantly reduce churn.

## **Business Recommendations**

Based on the findings, the following recommendations are proposed:

1. Encourage customers with month-to-month contracts to migrate toward annual or two-year plans through discounts, loyalty rewards, or bundled offers.
2. Conduct detailed customer satisfaction surveys among Fiber Optic subscribers to identify service quality issues and improve customer experience.
3. Review the Electronic Check payment process and encourage customers to adopt more convenient payment methods through promotional incentives.
4. Develop predictive churn models to proactively identify customers at high risk of leaving and implement personalized retention campaigns.
5. Continuously monitor customer behaviour through Power BI dashboards and update retention strategies based on evolving customer trends.

## **Conclusion**

This project successfully demonstrates how business intelligence tools can transform raw customer information into actionable business insights.

Through systematic data cleaning, DAX calculations, and interactive dashboard development, the project identified important customer churn patterns associated with contract type, payment method, internet service, and customer demographics.

The dashboard enables decision-makers to monitor customer retention performance efficiently and supports evidence-based business decisions aimed at reducing customer churn and improving long-term customer relationships.

The project also demonstrates practical skills in data preparation, business analytics, data visualization, and dashboard development using Microsoft Excel and Microsoft Power BI.

## Customer Churn Analysis

Contract

All

Internet Service

All

Payment Method

All

Gender

All

7K  
Total  
Customers

27%  
Churn Rate  
%

2K  
Churned  
Customers

5K  
Active  
Customers

\$64.76  
Average  
Charges  
(Monthly)

32.37  
Average  
Tenure  
(Monthly)

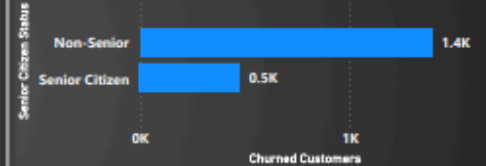
Churned Customers by PaymentMethod



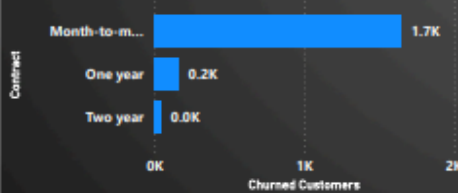
Customers by Gender



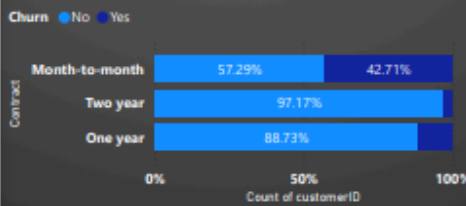
Churned Customers by Senior Citizen Status



Churned Customers by Contract



Customer Distribution by Contract



Churned Customers by InternetService

